

The Changing Value of K_W

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An brief exploration of how the water ion product, K_W , changes with temperature. We will also demonstrate the utility of the NIST Chemistry Webbook database in providing essential data for physical organic chemistry calculations.

Temperature and K_W

The ion product of water, K_W is famously stated to be 10^{-14} . In pure water. At 25 °C. Under one atmosphere. Chemistry is often performed under this standard condition, but is just as often performed at different temperatures or pressures.

The presence of solutes can change the value of K_W as well.¹ For example the value of K_W in seawater is $10^{-13.2}$ M. So, in neutral seawater we see that $[H_3O^+]$ and $[OH^-] = 10^{-6.6}$ M. The pH of neutral seawater is 6.6, not 7.0. This is important when calculating rates of specific base catalyzed reactions in pH -rate profiles. For example, at $pH = 7.0$ the concentration of hydroxide ion is $10^{-6.2}$ M, almost 10-fold greater than in neutral pure water.²

The K_W will change as conditions change. Pressure, temperature and ionic strength all affect the value.

In recent examples of pH rate profiles, we have seen reactions performed at temperatures other than 25 °C. In many cases one can find reported values for K_W at a given temperature but it is useful to be able to obtain a value at any temperature. The National Institute of Standards & Technology (NIST)³ provides a very useful resource, the Chemistry Webbook⁴ that contains many tables of physical and chemical properties.

The Marshall-Frank Model

The Marshall-Frank model describes the relationship between the value of K_W and temperature⁵ This paper reports an empirical formula for K_W vs. temperature. The expression is stated in eq. 1 with the parameters listed in table 1.

$$pK_W = A + \frac{B}{T} + \frac{C}{T^2} + \frac{D}{T^3} + \left[E + \frac{F}{T} + \frac{G}{T^2} \right] \log \rho \quad (1)$$

where ρ is the density of water in g/cm and the value of K_W is calculated in molal units (moles/kg of water).

This equation is an interpolation based on a polynomial fit through known values. The density correction component of the equation will need values of the density of pure water under standard pressure for each temperature.

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

¹ “The estimation of acid dissociation constants in seawater media from potentiometric titrations with strong base. I. The ionic product of water – K_W ,” A.G. Dickson, J.P. Riley, *Marine Chemistry*, 1979, 7, 89–99. [https://doi.org/10.1016/0304-4203\(79\)90001-X..](https://doi.org/10.1016/0304-4203(79)90001-X..)

² pH 7.0 does not mean "neutral", it means that $[H_3O^+] = 10^{-7}$ M. To be neutral, we must have a situation where $[H_3O^+] = [OH^-]$.

³ NIST is a division of the Department of Commerce of the USA federal government. <https://www.nist.gov>

⁴ <https://webbook.nist.gov>

⁵ “Ion product of water substance, 0–1000 °C, 1–10,000 bars New International Formulation and its background.” W.L. Marshall, E.U. Franck, *J. Phys. Chem. Ref. Data*, 1981, 10, 295–304. <https://doi.org/10.1063/1.555643>. Free access is available at the NIST at <https://srdata.nist.gov/jpcrd/jpcrd181.pdf>

Table 1: Parameters for the Marshall-Frank model⁵ for pK_W described in eq. 1. ↓

A	–4.098	
B	-3.2452×10^3	K
C	2.2362×10^5	K ²
D	-3.984×10^7	K ³
E	13.957	
F	-1.2623×10^3	K
G	8.5641×10^5	K ²

Density Tables

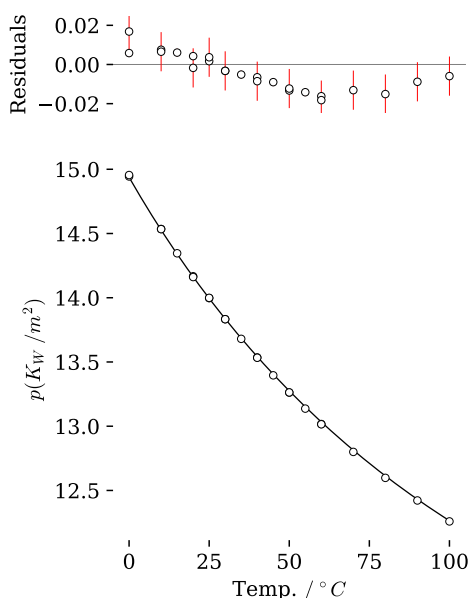
We can obtain a table of the density of water at temperatures between zero and 100 °C and then we can calculate the value of pK_W using the formula above. The NIST Chemistry Webbook will give a table of densities based on a database query. I queried values at unit degrees Celsius⁶ This set of values is available on the GitHub site.

We can also obtain a set of density values vs temperature from the CRC handbook of Chemistry and Physics.⁷ Sets of data obtained from the CRC handbook are available on the GitHub site.

When you examine the CRC and the NIST density tables you will see that they are identical within the limits of reported precision. They are very likely derived from the same literature sources. The field of chemistry has agreed on the density of water at standard pressure.

Calculating K_W

Using the density tables, we will calculate the value of K_W (in units of molality). We could convert from molality to molarity in pure water by converting the mass, one kg of water, to litres using the density table.⁸ A plot of calculated pK_W vs temperature is presented in figure 1.⁹ Experimental data^{10,11,13} is plotted along with the calculated values from the Marshall-Frank model.



⁶ “Thermophysical Properties of Fluid Systems”, E.W. Lemmon et al., *NIST Chemistry WebBook, NIST Standard Reference Database Number 69*, (retrieved September 28, 2024), P.J. Linstrom and W.G. Mallard, Eds., National Institute of Standards and Technology, Gaithersburg MD, 20899, <https://doi.org/10.18434/T4D303>.

⁷ Available via the UPEI Library at <https://library.upei.ca/chem>

⁸ You will see it makes almost no difference.

⁹ You can explore the *Python* notebook that was used to generate the plot and calculate values of K_W at a given temperature.

Figure 1: Plot of experimental data for pK_W vs temperature in pure water at standard pressure. The line is calculated using the Marshall-Frank model⁵ and parameters. Experimental data is from table 4 in reference 13. Residuals are the difference between the data and the model.

¹⁰ “A note on the temperature variation of the ionisation constants of weak electrolytes.” H.S. Harned, R.A. Robinson, *Trans. Faraday Soc.*, **1940**, *36*, 973-978. <https://doi.org/10.1039/TF9403600973>

¹¹ “Aussagen über die Eigendissoziation des Wassers aus Molwärmemessungen gelöster Elektrolyte.” T. Ackermann, *Z. Elektrochem., Ber. Bunsenges. Phys. Chem.*, **1958**, *62*, 411-419. <https://doi.org/10.1002/bbpc.19580620406>

IAPWS Model 2024

The Marshal-Frank model⁵ of 1981 fits the data very well. Although there is a systematic variation, it is almost insignificant. This model was part of an effort by the International Association for the Properties of Water and Steam (IAPWS) to provide tools for predicting the value of K_W in a wide range of conditions. It was the paper referenced in the NIST Chemistry Webbook. However, nothing stands still in science and the model has been improved¹² in 2006 and tweaked again¹³ in 2024.

The new IAPWS model is more complicated and it should give values very close to the 1981 model. Most of the improvements were for fitting data obtained in extreme conditions. The current IAPWS model is described in eqs. 2, 3 and 4 with the parameters listed in table 2. M_W is the molecular weight of water, 18.015 268 g mol⁻¹; $n = 6$ and is the ion coordination number for water; and pK_W^G is the gas phase ion product and represents a theoretical minimum value for pK_W at a given temperature.

$$pK_W = -2n \left[\log(1 + Z) - \frac{Z}{Z + 1} \rho \left(\beta_0 + \frac{\beta_1}{T} + \beta_2 \rho \right) \right] + pK_W^G + 2 \log \frac{M_W}{1000} \quad (2)$$

$$Z = \rho e^{\alpha_0 + \frac{\alpha_1}{T} + \frac{\alpha_2}{T^2} \rho^{\frac{2}{3}}} \quad (3)$$

$$pK_W^G = 0.61415 + \frac{4.825133 \times 10^4}{T} - \frac{6.770793 \times 10^4}{T^2} + \frac{1.0102100 \times 10^7}{T^3} \quad (4)$$

Using the density tables and the IAPWS model, we can calculate values for pK_W for any temperature at any pressure. The calculated values for standard pressure are presented in figure 2 on the next page along with experimental data. We see a better fit with no apparent systematic variations. The deviation of the Marshal-Frank model from the IAPWS 2024 model is plotted in the residual plot. Although it does deviate, the difference is very small.

Roll Your Own Model

The Marshall-Frank model and the IAPWS model are merely polynomial fits to the data. Even the IAPWS model, with its three layers of polynomial fits, is just such a sophisticated interpolation. Neither is based on any fundamental theoretical principles; The complexity of the IAWPS model matches the

¹² "The Ionization Constant of Water over Wide Ranges of Temperature and Density." A.V. Bandura, S.N. Lvov, *J. Phys. Chem. Ref. Data*, 2006, 35, 15-30. <https://doi.org/10.1063/1.1928231>

¹³ "Revised Parameters for the IAPWS Formulation for the Ionization Constant of Water Over a Wide Range of Temperatures and Densities, Including Near-Critical Conditions." H. Arcis, M. Bachet, et al., *J. Phys. Chem. Ref. Data*, 2024, 53, 023103. <https://doi.org/10.1063/5.0198792>

Table 2: Fitted parameters of the 2024 IAPWS model¹³ for pK_W described in eqs. 2, 3 and 4. ↓

α_0	-0.702132	
α_1	8681.05	K
α_2	-24145.1	$K^2(g\text{ cm}^{-3})^{-2/3}$
β_0	0.813876	$(g\text{ cm}^{-3})^{-1}$
β_1	-51.4471	$K(g\text{ cm}^{-3})^{-1}$
β_2	-0.469920	$(g\text{ cm}^{-3})^{-2}$

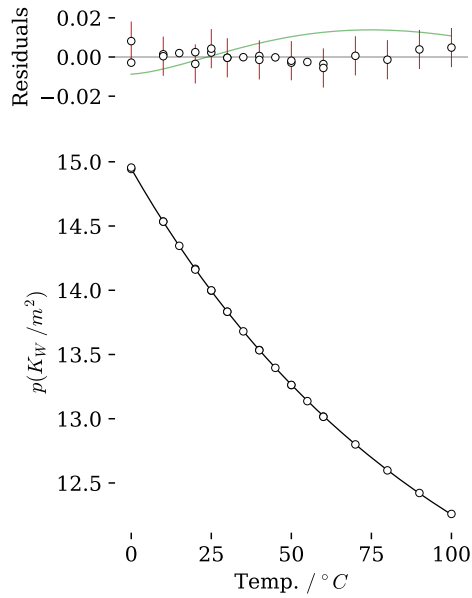


Figure 2: Plot of experimental data for pK_W vs temperature in pure water at standard pressure. The line is calculated using the 2024 IAPWS model and parameters¹³. Experimental data is from table 4 in reference 13. Residuals are the difference between the data and the model. In the residual plot, the green line is the difference between the Marshall-Frank model⁵ and the IAPWS model.

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measured values for pK_W in all conditions except ice. The Marshall-Frank model only works for water in liquid phase. They are almost perfectly co-linear in the tame world of water at standard pressure.

Just for fun, let us curve fit the experimental data that we have using a simple polynomial interpolation. Will it be as good as the professional models within the limited range of temperatures that we care about?

In Figure 3 on the following page is presented the experimental data and the line calculated from the polynomial fitting. The difference between the Marshall-Frank model and the IAPWS model from the polynomial fits is plotted on the residual plot for comparison. How did we do?

Comments and Conclusions

Both the Marshall-Frank model and the IAPWS 2024 models fit the experimental data very well. The IAPWS model was superior, however a simple polynomial fit to the experimental data produced an almost identical fit when pressure was set a 1 atm.

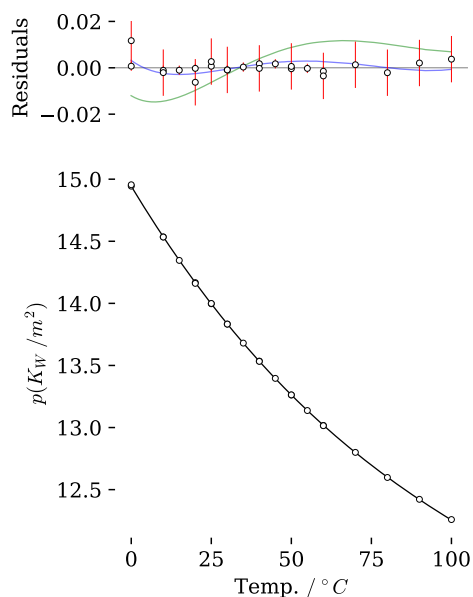


Figure 3: Plot of experimental data for pK_W vs temperature in pure water at standard pressure. The line is calculated using a 3rd-degree polynomial curve fit. Experimental data is from table 4 in reference 13. Residuals are the difference between the data and the polynomial fit. In the residual plot, the green line is the difference between the polynomial fit and the Marshall-Frank model⁵ and the blue line is the deviation of the IAPWS 2024 model.¹³

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Notes and Supplementary material

Python Notebook

The calculations, plots and data processing is documented in the *Python* notebook that accompanies this document. The data is available as text files (both csv and tab-delimited) and are also available. All files are available via the *GitHub* repository.

A Note of Pressure and Density

Both the Marshall-Frank model and the 2024 IAPWS model use the density of water rather than pressure in the equations. The pressure is whatever crushed the water to the measured density. When working with liquid water in a reactor at 600 °C, we know the mass and we know the volume. So, we will know the density at whatever pressure is in that vessel.¹⁴ The density will always be close to 1.0 unless we are dealing with steam or supercritical water. These models are designed to fit to experimental data in all these extreme cases while also giving accurate results at STP.

¹⁴ Hopefully it was designed under the supervision of an experienced professional engineer. Everyone in the room will be killed if it explodes. Do not work with uncertified pressure vessels.

The Shifting Sands of the SI System

You may have noticed that the density of water never reached 1.0000 g/cm³ in the density tables. Perhaps you expected that the density should be exactly one kilogram per litre at 4 °C? That was the original definition in the SI system as established by the utopian dreamers of the French revolution.

However, the Treaty of the Metre, signed in 1875 established a set of standards for the meter and the kilogram based upon a rod of standard length and a cylinder of standard mass. Since the definition of the gram and of the centimetre changed just a little bit on that day, we have not had a density for water of exactly one ever again.¹⁵

Forty cylinders of platinum-iridium alloy were manufactured as standard kilograms according to the treaty and were distributed to member nations as national standards. One cylinder was kept by the International Bureau of Weights and Measures as the international standard against which all others would be standardized.¹⁶ The cylinders have been compared twice over the last century and their masses were diverging due to handling and wear. In 2018, the kilogram was redefined again based on fundamental physical constants of the universe. Now those forty cylinders are just collectibles. I want one.

Experimental Data for pK_W

The data in table 3 on the following page is available as a csv file. It is presented here so that this document can stand alone in the absence of the supplementary files. With some knowledge of *Python* and the mathematical models described in this document the reader should be able to reproduce the plots and analysis.

A Self-Contained Model

Both the Marshal-Frank model and the IAPWS 2024 model required density data for water. Tables of density vs temperature at standard pressure are available and were used in the calculations demonstrated above.

The models are designed to give accurate values in a large range of temperatures and pressures. If we limit ourselves to atmospheric pressure we could fit the data to a polynomial function and create an empirical model for the density of water at a given temperature. In figure 4 on the following page is plotted the density data for water from the CRC Handbook vs temperature with a 4th-degree polynomial fit. The residuals for that fit against the data are plotted along with the residuals from a 3rd-degree polynomial fit. Both gave excellent results with only very small deviations from the data. However I chose to use the 4th-degree polynomial fit described in eq. 5.

$$\rho = -1.31735 + 2.625340 \times 10^{-2}T - 1.102577 \times 10^{-4}T^2 + 2.057821 \times 10^{-07}T^3 - 1.475348 \times 10^{-10}T^4 \quad (5)$$

If we have a temperature between zero and 100 °C at standard pressure, we can calculate a value for the density of water. Then we can use this value in either model for pK_W . This eliminates the need for a table of density data.

¹⁵ When one is exploring a subject, sometimes one goes down a rabbit hole. The changing definitions of the meter and the gram became my rabbit hole once I observed that the maximum density of water was not "one" by definition. I followed the trail using Wikipedia. You can start on this journey for yourself at https://en.wikipedia.org/wiki/2019_revision_of_the_SI

¹⁶ "One Cylinder to rule them all,
One Cylinder to find them,
One Cylinder to bring them all,
and in the city of Paris bind them."

Temp /($^{\circ}\text{C}$)	ρ /(g cm $^{-3}$)	pK_W /(m 2)	Std. Err. /(m 2)
0.0	0.9998	14.944	0.001
10.0	0.9997	14.535	0.001
15.0	0.9991	14.346	0.001
20.0	0.9982	14.167	0.001
25.0	0.9970	13.997	0.001
30.0	0.9956	13.833	0.001
35.0	0.9940	13.680	0.001
40.0	0.9922	13.535	0.001
45.0	0.9902	13.396	0.001
50.0	0.9880	13.262	0.001
55.0	0.9857	13.137	0.001
60.0	0.9832	13.017	0.001
0.0	0.9998	14.955	0.005
10.0	0.9997	14.534	0.005
20.0	0.9982	14.161	0.005
25.0	0.9970	13.999	0.005
30.0	0.9956	13.833	0.005
40.0	0.9922	13.533	0.005
50.0	0.9880	13.263	0.005
60.0	0.9832	13.015	0.005
70.0	0.9778	12.800	0.005
80.0	0.9718	12.598	0.005
90.0	0.9653	12.422	0.005
100.0	0.9583	12.259	0.005

Table 3: Experimental values selected from Table 4 of the AIST 2024 report.¹³

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Table 4 reports many data sets, many representing extreme conditions. Data sets from Harned & Robinson,^{??} and from Ackermann,^{??} which were measured at standard pressure, are reported here.

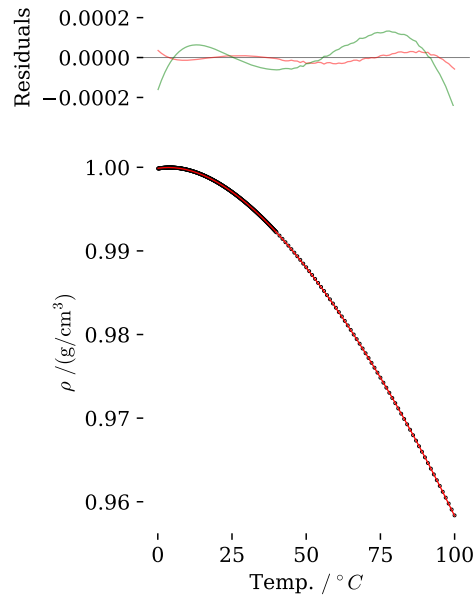


Figure 4: Plot of CRC Handbook data for density vs temperature in pure water at standard pressure. The line is calculated using a 4th-degree polynomial curve fit. In the residual plot, the red line is the difference between the 4th-degree polynomial fit and the data; the green line is the deviation for a 3rd-degree polynomial fit.

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